# Data Sheet

## EXPERIMENT NO. 11B

**Title: -** Simulation of Monostable Multivibrator using IC 555 (using Tinkercad / Proteus software circuit).

## Instrument & Component Required: -

| Sl. No. | Apparatus / Component | Specification | Quantity |
|---|---|---|---|
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## Important Formulae: -

## Operating Principle: -

The monostable (one-shot) multivibrator has one stable state. A negative trigger pulse at pin 2 (below $V_{CC}$ /3) switches the output HIGH for a fixed time T, after which it automatically returns to the stable LOW state.

## Output Pulse Width: -

$T = 1.1 \times R \times C$

- R = Timing resistor (connected between pin 7 and $V_{CC}$)
- C = Timing capacitor (connected between pin 6/7 and ground)
- The capacitor charges through R until it reaches (2/3) $V_{CC}$, at which point the output resets to LOW

## Observation Table: -

| Sl. No. | R (kΩ) | C (µF) | Pulse Width, T (Theoretical) ms | Pulse Width, T (Practical / Measured) ms |
|---|---|---|---|---|
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## Calculation: -

T (Theoretical) = $1.1 \times R \times C$ = _____

% Error between Theoretical and Practical Pulse Width = _____ %

**Conclusion: -**

**Teacher's Sign and Date**

# Data Sheet

## EXPERIMENT NO. 11A

**Title: -** Simulation of Astable Multivibrator using IC 555 (using Tinkercad / Proteus software circuit).

## Instrument & Component Required: -

| Sl. No. | Apparatus / Component | Specification | Quantity |
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## Important Formulae: -

## Operating Principle: -

The astable multivibrator has no stable state — it free-runs, continuously switching between HIGH and LOW, generating a rectangular wave at the output (pin 3) without any external trigger.

## Charging & Discharging Time: -

Charging time (output HIGH), $t_1 = 0.693 \times (R_A + R_B) \times C$

Discharging time (output LOW), $t_2 = 0.693 \times R_B \times C$

## Time Period & Frequency: -

$T = t_1 + t_2 = 0.693 \times (R_A + 2R_B) \times C$

$f = 1 / T = 1.44 / [(R_A + 2R_B) \times C]$

## Duty Cycle: -

% Duty Cycle $= [(R_A + R_B) / (R_A + 2R_B)] \times 100$

- $R_A$ = Resistor between $V_{CC}$ and pin 7
- $R_B$ = Resistor between pin 7 and pin 2/6
- C = Timing capacitor between pin 2/6 and ground

## Observation Table: -

| Sl. No. | $R_A$ (kΩ) | $R_B$ (kΩ) | C (µF) | f (Theoretical) Hz | f (Practical / Measured) Hz | Duty Cycle (%) |
|---|---|---|---|---|---|---|
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| Sl. No. | $R_A$ (kΩ) | $R_B$ (kΩ) | C (µF) | f (Theoretical) Hz | f (Practical / Measured) Hz | Duty Cycle (%) |
|---|---|---|---|---|---|---|
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## Calculation: -

f (Theoretical) = $1.44 / [(R_A + 2R_B) \times C]$ = _____

Duty Cycle = $[(R_A + R_B) / (R_A + 2R_B)] \times 100$ = _____ %

% Error between Theoretical and Practical Frequency = _____ %

## Conclusion: -

**Teacher's Sign and Date**

# Data Sheet

## EXPERIMENT NO. 12

**Title: -** Simulation of Schmitt Trigger Circuit using IC 555 (using Tinkercad / Proteus software circuit).

## Instrument & Component Required: -

| Sl. No. | Apparatus / Component | Specification | Quantity |
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## Important Formulae: -

## Operating Principle: -

Pins 2 and 6 of the IC 555 are tied together and driven by the input signal. The internal comparators (with reference levels at $V_{CC}$ /3 and 2 $V_{CC}$ /3) convert a slowly-varying input into a clean rectangular output with two well-defined switching thresholds, giving the circuit hysteresis.

## Upper & Lower Threshold Voltages: -

Upper Threshold Voltage, $V_{UT} = (2/3) \times V_{CC}$

Lower Threshold Voltage, $V_{LT} = (1/3) \times V_{CC}$

## Hysteresis Width: -

$V_H = V_{UT} - V_{LT} = V_{CC} / 3$

- Output switches LOW → HIGH when Vin falls below $V_{LT}$
- Output switches HIGH → LOW when Vin rises above $V_{UT}$

## Observation Table: -

| Sl. No. | $V_{CC}$ (V) | $V_{UT}$ (Theoretical) (V) | $V_{LT}$ (Theoretical) (V) | $V_{UT}$ (Measured) (V) | $V_{LT}$ (Measured) (V) |
|---|---|---|---|---|---|
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## Calculation: -

$V_{UT}$ (Theoretical) = (2/3) × $V_{CC}$ = _____

$V_{LT}$ (Theoretical) = (1/3) × $V_{CC}$ = _____

Hysteresis, $V_H = V_{UT} - V_{LT}$ = _____

## Conclusion: -

**Teacher's Sign and Date**